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```
**LAB 18.2: Sequence-to-Sequence Tasks — Using Hugging Face Pipe
Summarization**

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```

LAB 18.2: Sequence-to-Sequence Tasks — Using Hugging Face Pipelines for Summarization

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STEP 1 — Install and Import Libraries

```
# @title STEP 1 — Install and Import Libraries
# Install required libraries
!pip install transformers torch

# Import libraries
from transformers import pipeline

print("Libraries installed and imported successfully!")
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transform
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->h
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->h
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3
Libraries installed and imported successfully!
```

STEP 2 — Load Summarization Model

```
# @title STEP 2 — Load Summarization Model
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Load tokenizer and model explicitly
tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")

# Define a summarization function using the loaded model and tokenizer
def summarizer(text, max_length=150, min_length=40, do_sample=False):
    inputs = tokenizer(text, max_length=1024, return_tensors="pt", truncation=True)
    summary_ids = model.generate(inputs["input_ids"], num_beams=4, max_length=max_length, min_length=min_length, early_stop
    return [tokenizer.decode(g, skip_special_tokens=True, clean_up_tokenization_spaces=False) for g in summary_ids]

print("Summarization model and tokenizer loaded successfully!")
```

```
vocab.json:         899k/?  [00:00<00:00,  8.96MB/s]

merges.txt:         456k/?  [00:00<00:00,  7.15MB/s]

tokenizer.json:         1.36M/?  [00:00<00:00,  20.8MB/s]

model.safetensors:  100%                                                    1.63G/1.63G  [00:14<00:00,  145MB/s]
Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights:  100%                          511/511  [00:00<00:00,  648.18it/s,  Materializing param=model.encoder.layers.11.self_attn_layer_norm.weight]

generation_config.json:  100%                                                    363/363  [00:00<00:00,  15.6kB/s]
Summarization model and tokenizer loaded successfully!
```

STEP 3 — Prepare Input Texts (Dataset)

```
# @title STEP 3 — Prepare Input Texts (Dataset)
# Dataset with different text types

news_text = """India successfully launched a new satellite into orbit today, marking a significant milestone
in its space exploration program. The mission aims to enhance communication systems and provide
better weather forecasting capabilities."""

technical_text = """Machine learning is a subset of artificial intelligence that focuses on building systems that
learn from data. Algorithms improve automatically through experience without being explicitly programmed."""

narrative_text = """Once upon a time, there was a young boy who dreamed of becoming a great scientist. Despite facing
many challenges, he remained determined and continued his studies."""

blog_text = """Maintaining a healthy lifestyle requires a balance between proper diet, regular exercise, and mental well-be
Small daily habits can greatly improve overall health."""

conversation_text = """Hey, did you finish the assignment? Not yet, I was planning to complete it tonight.
The deadline is tomorrow, so we should hurry."""

texts = [news_text, technical_text, narrative_text, blog_text, conversation_text]

print("Dataset prepared!")
```

```
Dataset prepared!
```

STEP 4 — Generate Summaries

```
# @title STEP 4 — Generate Summaries
# Generate summaries

for i, text in enumerate(texts):
    summary = summarizer(text, max_length=60, min_length=20, do_sample=False)
    print(f"\nText {i+1} Summary:")
    print(summary[0])
```

```
Text 1 Summary:
India successfully launched a new satellite into orbit today. Mission aims to enhance communication systems and provide bett

Text 2 Summary:
Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data. Algorithms im

Text 3 Summary:
Once upon a time, there was a young boy who dreamed of becoming a great scientist. Despite facing many challenges, he remair

Text 4 Summary:
Maintaining a healthy lifestyle requires a balance between proper diet, regular exercise, and mental well-being. Small daily

Text 5 Summary:
Hey, did you finish the assignment? Not yet, I was planning to complete it tonight. The deadline is tomorrow, so we should h
```

STEP 5 — Control Summary Length

```
# @title STEP 5 — Control Summary Length
sample_text = texts[0]

# Short Summary
short_summary = summarizer(sample_text, max_length=30, min_length=10, do_sample=False)
print("\nShort Summary:")
print(short_summary[0])

# Medium Summary
medium_summary = summarizer(sample_text, max_length=60, min_length=20, do_sample=False)
```

```
print("\nMedium Summary:")
print(medium_summary[0])

# Long Summary
long_summary = summarizer(sample_text, max_length=100, min_length=40, do_sample=False)
print("\nLong Summary:")
print(long_summary[0])
```

```
Short Summary:
India successfully launched a new satellite into orbit today. Mission aims to enhance communication systems and provide bett

Medium Summary:
India successfully launched a new satellite into orbit today. Mission aims to enhance communication systems and provide bett

Long Summary:
India successfully launched a new satellite into orbit today. Mission aims to enhance communication systems and provide bett
```

STEP 6 — Compare Text Types

```
# @title STEP 6 — Compare Text Types
# Simple comparison display

for i, text in enumerate(texts):
    summary = summarizer(text, max_length=50, min_length=15, do_sample=False)
    print(f"\n--- Text Type {i+1} ---")
    print("Original:", text)
    print("Summary:", summary[0])
```

```
--- Text Type 1 ---
Original: India successfully launched a new satellite into orbit today, marking a significant milestone
in its space exploration program. The mission aims to enhance communication systems and provide
better weather forecasting capabilities.
Summary: India successfully launched a new satellite into orbit today. Mission aims to enhance communication systems and pro

--- Text Type 2 ---
Original: Machine learning is a subset of artificial intelligence that focuses on building systems that
learn from data. Algorithms improve automatically through experience without being explicitly programmed.
Summary: Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data. Algo

--- Text Type 3 ---
Original: Once upon a time, there was a young boy who dreamed of becoming a great scientist. Despite facing
many challenges, he remained determined and continued his studies.
Summary: Once upon a time, there was a young boy who dreamed of becoming a great scientist. Despite facing many challenges,

--- Text Type 4 ---
Original: Maintaining a healthy lifestyle requires a balance between proper diet, regular exercise, and mental well-being.
Small daily habits can greatly improve overall health.
Summary: Maintaining a healthy lifestyle requires a balance between proper diet, regular exercise, and mental well-being. Sm

--- Text Type 5 ---
Original: Hey, did you finish the assignment? Not yet, I was planning to complete it tonight.
The deadline is tomorrow, so we should hurry.
Summary: Hey, did you finish the assignment? Not yet, I was planning to complete it tonight. The deadline is tomorrow, so we
```

STEP 7 — Very Long Text Test

```
# @title STEP 7 — Very Long Text Test
long_text = """Artificial Intelligence has become one of the most transformative technologies of the modern era.
It is used in healthcare for diagnosing diseases, in finance for fraud detection, and in transportation for autonomous vehi
Despite its advantages, AI raises ethical concerns such as job displacement and privacy issues.
Researchers are working to improve transparency and fairness."""

summary = summarizer(long_text, max_length=80, min_length=30, do_sample=False)

print("\nLong Text Summary:")
print(summary[0])
```

```
Long Text Summary:
Artificial Intelligence has become one of the most transformative technologies of the modern era. Despite its advantages, AI
```

STEP 8 — Generate Lab Report (CODE)

```
# @title STEP 8 — Generate Lab Report (CODE)
# STEP 8 — Generate Lab Report Automatically

report = ""

# Objective
```

```
report += "OBJECTIVE:\n"
report += "To perform text summarization using Hugging Face Transformers and analyze the results.\n\n"

# Dataset Description
report += "DATASET DESCRIPTION:\n"
report += "The dataset consists of multiple text passages from different domains such as news, technical, narrative, blog, a

# Summarization Outputs
report += "SUMMARIZATION OUTPUTS:\n"

for i, text in enumerate(texts):
    summary = summarizer(text, max_length=60, min_length=20, do_sample=False)
    report += f"\nText {i+1}:\n"
    report += f"Original: {text}\n"
    report += f"Summary: {summary[0]}\n"

# Parameter Comparison
report += "\nPARAMETER ANALYSIS:\n"

short_summary = summarizer(texts[0], max_length=30, min_length=10, do_sample=False)
medium_summary = summarizer(texts[0], max_length=60, min_length=20, do_sample=False)
long_summary = summarizer(texts[0], max_length=100, min_length=40, do_sample=False)

report += f"\nShort Summary: {short_summary[0]}\n"
report += f"\nMedium Summary: {medium_summary[0]}\n"
report += f"\nLong Summary: {long_summary[0]}\n"

# Analysis
report += "\nANALYSIS:\n"
report += "The model performs well on structured text like news but loses some details in conversational text. Summary quali

# Conclusion
report += "\nCONCLUSION:\n"
report += "Hugging Face models provide efficient summarization but have limitations with long and informal texts.\n"

# Save report to file
with open("Lab18_Report.txt", "w") as file:
    file.write(report)

print("Report generated successfully as Lab18_Report.txt")
```

```
Report generated successfully as Lab18_Report.txt
```